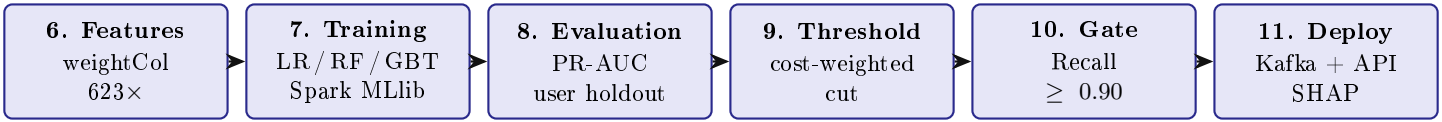
6. Features
weightCol
623×
7. Training
LR / RF / GBT
Spark MLlib
8. Evaluation
PR-AUC
user holdout
9. Threshold
cost-weighted
cut
10. Gate
Recall
≥ 0.90
11. Deploy
Kafka + API
SHAP